

Schrödinger Bridge for Non-linear ROMs

Presentation of Progress

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Problem Setup and Notation

Let $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ be two probability measures (initial and terminal marginals), time rescaled to $[0, 1]$, $\gamma > 0$ the diffusion coefficient.

We seek a time-dependent density–velocity pair $(\rho_t, b_t)_{t \in [0, 1]}$ such that:

- $\rho_0 = \mu, \rho_1 = \nu$
- ρ_t evolves according to the **Fokker–Planck equation**

$$\partial_t \rho_t + \nabla \cdot (\rho_t b_t) = \gamma \Delta \rho_t$$

- The pair minimizes the **expected kinetic energy**

$$\mathcal{J}(\rho, b) = \int_0^1 \int_{\mathbb{R}^d} \frac{1}{2} |b_t(x)|^2 \rho_t(x) dx dt$$

This arises from Girsanov's theorem applied to the controlled diffusion

$$dX_t = b_t(X_t) dt + \sqrt{2\gamma} dW_t,$$

where the relative entropy w.r.t. Brownian motion is $\text{KL}(\mathbb{P} \parallel \mathbb{R}) = \mathcal{J}/(2\gamma)$.

The Constrained Variational Problem

The Schrödinger bridge is the solution to

$$\inf_{(\rho, b)} \mathcal{J}(\rho, b) \quad \text{subject to} \quad \partial_t \rho_t + \nabla \cdot (\rho_t b_t) = \gamma \Delta \rho_t, \quad \rho_0 = \mu, \quad \rho_1 = \nu.$$

Introduce a Lagrange multiplier $\phi \in C^1([0, 1] \times \mathbb{R}^d)$ to enforce the constraint. The augmented Lagrangian reads:

$$\mathcal{L}(\rho, b, \phi) = \int_0^1 \int_{\mathbb{R}^d} \left(\frac{1}{2} |b_t|^2 \rho_t - \phi_t (\partial_t \rho_t + \nabla \cdot (\rho_t b_t) - \gamma \Delta \rho_t) \right) dx dt.$$

Stationarity $\delta \mathcal{L} = 0$ with respect to ρ , b , and ϕ yields the Euler–Lagrange system.

Euler–Lagrange Optimality Conditions

Variation w.r.t. $b \Rightarrow$ optimal drift

$$b_t(x) = \nabla \phi_t(x).$$

Variation w.r.t. $\rho \Rightarrow$ Hamilton–Jacobi–Bellman equation (backward)

$$\partial_t \phi_t + \frac{1}{2} |\nabla \phi_t|^2 + \gamma \Delta \phi_t = 0.$$

Coupled Euler–Lagrange system

$$\begin{cases} \partial_t \phi_t + \frac{1}{2} |\nabla \phi_t|^2 + \gamma \Delta \phi_t = 0, \\ \partial_t \rho_t + \nabla \cdot (\rho_t \nabla \phi_t) = \gamma \Delta \rho_t, \\ \rho_0 = \mu, \quad \rho_1 = \nu. \end{cases}$$

Hopf–Cole Transformation (1/2) — Linearization

Backward wave function

Define η_t^* via the value function ϕ_t :

$$\eta_t^*(x) = \exp\left(\frac{\phi_t(x)}{2\gamma}\right).$$

Substituting into the HJB equation gives the **backward heat equation**:

$$\partial_t \eta_t^* = -\gamma \Delta \eta_t^*.$$

Forward wave function

Define η_t such that $\rho_t(x) = \eta_t(x) \eta_t^*(x)$. Substituting into the Fokker–Planck equation (using $b = 2\gamma \nabla \log \eta^*$) gives the **forward heat equation**:

$$\partial_t \eta_t = \gamma \Delta \eta_t.$$

Hopf–Cole Transformation (2/2) — Results

Schrödinger system (boundary conditions in η -variables)

$$\eta_0(x) \eta_0^*(x) = \mu(x), \quad \eta_1(x) \eta_1^*(x) = \nu(x).$$

Optimal drift

$$b_t(x) = 2\gamma \nabla \log \eta_t^*(x).$$

Neumann boundary conditions on bounded domains Ω

Zero-flux $J_t \cdot n = (\rho_t b_t - \gamma \nabla \rho_t) \cdot n = 0$ on $\partial\Omega$ reduces to:

$$\gamma \left(\eta_t \frac{\partial \eta_t^*}{\partial n} - \eta_t^* \frac{\partial \eta_t}{\partial n} \right) = 0 \quad \implies \quad \frac{\partial \eta_t}{\partial n} = 0, \quad \frac{\partial \eta_t^*}{\partial n} = 0 \quad \text{on } \partial\Omega.$$

Both linear heat equations are solved with **homogeneous Neumann BCs**.

Log-Potential Formulation

Definition

Define the forward and backward log-potentials:

$$f_t(x) := \log \eta_t(x), \quad g_t(x) := \log \eta_t^*(x).$$

Density, drift, and marginal constraints

$$\rho_t(x) = e^{f_t(x) + g_t(x)}, \quad b_t(x) = 2\gamma \nabla g_t(x).$$

$$f_0 + g_0 = \log \mu, \quad f_1 + g_1 = \log \nu.$$

Remark

The heat equations for (η, η^*) translate to nonlinear parabolic equations for (f, g) that are *not* solved directly. Instead, IPFP alternates linear heat solves and logarithmic rescaling of boundary conditions.

IPFP in Log-Potential Form

Let $\mathcal{P}_t$ denote the heat semigroup generated by $\gamma\Delta$ with Neumann BCs.

IPFP iterations (Sinkhorn in log-form)

$$g^{(k+1)}(x) = \log \nu(x) - \log\left(\mathcal{P}_1[e^{f^{(k)}}](x)\right),$$

$$f^{(k+1)}(x) = \log \mu(x) - \log\left(\mathcal{P}_1[e^{g^{(k+1)}}](x)\right).$$

Recovery of the optimal flow

Once convergence is reached:

$$\rho_t(x) = \exp(f_t(x) + g_t(x)), \quad b_t(x) = 2\gamma \nabla g_t(x).$$

The pair (ρ, b) solves the Euler–Lagrange system and minimizes $\mathcal{J}(\rho, b)$.

The Heat Semigroup $\mathcal{P}_t$

$\mathcal{P}_t$ denotes the semigroup generated by $\gamma\Delta$ with Neumann BCs on Ω . On $\mathbb{R}^d$ it has an explicit kernel:

$$(\mathcal{P}_t f)(x) = \int_{\mathbb{R}^d} K_t(x, y) f(y) dy, \quad K_t(x, y) = \frac{1}{(4\pi\gamma t)^{d/2}} \exp\left(-\frac{\|x - y\|^2}{4\gamma t}\right).$$

Key identity: Gibbs kernel $\equiv$ heat semigroup

The Sinkhorn kernel $K_{ij} = e^{-\|x_i - x_j\|^2/\varepsilon}$ with $\varepsilon = 4\gamma$ is *exactly* the heat semigroup $e^{\gamma\Delta}$ evaluated at $t = 1$. This means the same FVM solver used inside IPFP also computes the convolutional Sinkhorn kernel, which we reuse for the W_2 error metric.

On bounded domains

No closed-form kernel is available. We solve $\partial_t \varphi = \gamma\Delta\varphi$ numerically on the unstructured aerodynamic mesh using a Finite Volume Method with homogeneous Neumann boundary conditions.

Two-Point Flux Approximation (TPFA)

Flux across face e_{ij} separating cells K_i and K_j :

$$F_{ij} = -\gamma \frac{\varphi_j - \varphi_i}{\|x_j - x_i\|} |e_{ij}|.$$

Neumann BC: ghost-cell reflection $\varphi_j = \varphi_i$ on $\partial\Omega \Rightarrow$ zero flux. Semi-discrete:

$$\mathcal{L}(\varphi)_i = |K_i|^{-1} \sum_{j \sim i} F_{ij}.$$

SSP-RK2 (Heun's method)

$$\varphi^* = \varphi^n + \Delta t \mathcal{L}(\varphi^n), \quad \varphi^{n+1} = \frac{1}{2}(\varphi^n + \varphi^* + \Delta t \mathcal{L}(\varphi^*)).$$

CFL: $\Delta t \leq \text{CFL} \cdot h^2 / \gamma$, $h = 10^{\text{th}}$ -percentile cell size (avoids worst-cell control). Step count n is a compile-time static constant (`jax.lax.fori_loop` fusion).

Anderson Acceleration of IPFP

IPFP is the fixed-point iteration $x^{(k+1)} = F(x^{(k)})$, $x = \text{vec}(f, g) \in \mathbb{R}^{2N}$, with linear rate $\rho < 1$.

Anderson(m) extrapolation

Store the last m proposals $X = [x^{(k-m+1)}, \dots, x^{(k)}]$ and residuals $R = [r^{(k-m+1)}, \dots, r^{(k)}]$, $r^{(j)} = F(x^{(j)}) - x^{(j)}$. Find the convex combination minimising the residual norm:

$$\min_{\substack{c \in \mathbb{R}^m \\ \mathbf{1}^T c = 1}} \|Rc\|_2^2 \implies c = \frac{(R^T R)^{-1} \mathbf{1}}{\mathbf{1}^T (R^T R)^{-1} \mathbf{1}}, \quad x^{(k+1)} = Xc.$$

Cost and convergence

Each step: one Sinkhorn step (2 heat solves) + $\mathcal{O}(m^2 N)$ least-squares (negligible).

Anderson(5): ~ 150 – 160 measured iterations to tol 10^{-7} vs. ~ 2150 for plain Sinkhorn ($> 13\times$ faster).

Reference Diffusion with a Drift

Replace Brownian motion by a reference diffusion carrying a prescribed macroscopic drift $\mathbf{b}_t(\mathbf{x})$,

$$dX_t = \mathbf{b}_t(X_t) dt + \sqrt{2\gamma} dW_t,$$

and penalize deviation of the control from this reference:

$$\mathcal{J}(\rho, b) = \int_0^1 \int_{\mathbb{R}^d} \frac{1}{2} \|b_t(x) - \mathbf{b}_t(x)\|^2 \rho_t(x) dx dt.$$

Optimal control and advected HJB equation

Varying the Lagrangian in b and then in ρ gives

$$b_t = \mathbf{b}_t + \nabla \phi_t, \quad \partial_t \phi_t + \frac{1}{2} \|\nabla \phi_t\|^2 + \mathbf{b}_t \cdot \nabla \phi_t + \gamma \Delta \phi_t = 0.$$

Idea: if $\mathbf{b}$ already carries the shock structure from M_0 toward M_1 , the residual correction $\nabla \phi_t$ stays small — the bridge only has to fix up what the drift misses, so it stays well-conditioned at small γ .

The Two Transport–Diffusion Equations

Hopf–Cole linearization of the drifted HJB equation gives a **backward** equation for η_t^* and a **forward** equation for η_t (with $\rho_t = \eta_t \eta_t^*$); together they replace the two heat equations (??)–(??) of the driftless bridge.

Backward — non-conservative advection

$$\partial_t \eta_t^* + \mathbf{b}_t \cdot \nabla \eta_t^* + \gamma \Delta \eta_t^* = 0.$$

Forward — conservative advection (Proposition, adjoint generator)

$$\partial_t \eta_t + \nabla \cdot (\mathbf{b}_t \eta_t) - \gamma \Delta \eta_t = 0, \quad \text{i.e.} \quad \partial_t \eta_t + \mathbf{b}_t \cdot \nabla \eta_t + (\nabla \cdot \mathbf{b}_t) \eta_t - \gamma \Delta \eta_t = 0.$$

Why two distinct operators

Without drift, the heat semigroup is self-adjoint, so one FVM solver serves both Sinkhorn half-steps. A drift breaks self-adjointness: the backward equation transports $\mathbf{b} \cdot \nabla \eta^*$ (non-conservative), the forward one transports $\nabla \cdot (\mathbf{b} \eta)$ (conservative) — they differ by the

IPFP with a Reference Drift

Let $Q_\tau = e^{\tau\mathcal{L}}$, $\mathcal{L} = \mathbf{b} \cdot \nabla + \gamma\Delta$ be the (non-conservative) transition operator of the drifted reference SDE, and $Q_\tau^\dagger = e^{\tau\mathcal{L}^*}$ its adjoint (conservative).

Drifted IPFP iterations

$$f^{(k+1)} = \log \mu - \log Q_1[e^{g^{(k)}}], \quad g^{(k+1)} = \log \nu - \log Q_1^\dagger[e^{f^{(k+1)}}].$$

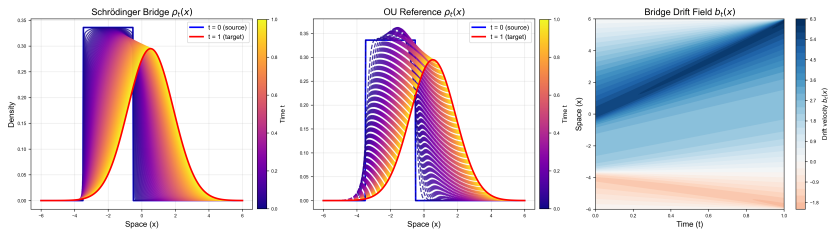
Discretization

$Q_1, Q_1^\dagger$ are built on the same TPFA diffusion stencil as $\mathcal{P}_1$, with a first-order-upwind advective flux; Q is assembled as the *exact area-weighted transpose* of $Q^\dagger$'s conservative upwind flux, guaranteeing an M-matrix (unconditional positivity, any Péclet number) and machine-precision adjointness $\langle Qu, v \rangle = \langle u, Q^\dagger v \rangle$.

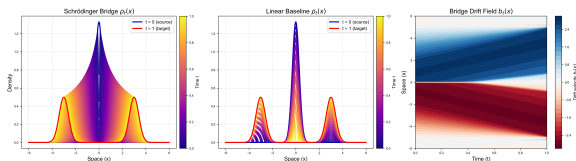
Consistency check

$\mathbf{b} \equiv 0 \Rightarrow Q_1 = Q_1^\dagger = \mathcal{P}_1$, and the drifted IPFP reduces exactly to the driftless one.

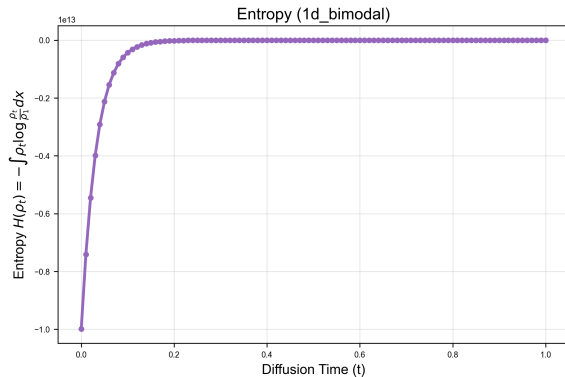
1D Validation: Square $\rightarrow$ OU Process ($\gamma = 0.05$)



1D Validation: Gaussian $\rightarrow$ Bimodal ($\gamma = 0.05$)



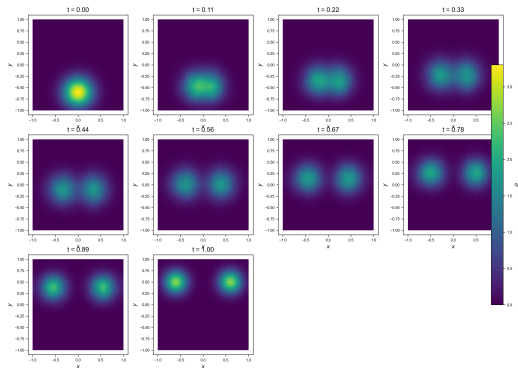
Density evolution (splitting transport)



Entropy $H[\rho_t]$ along the bridge

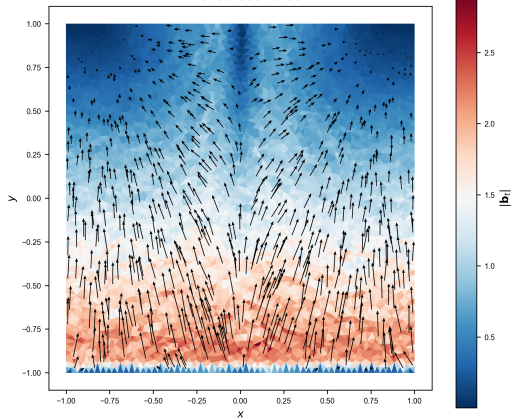
2D Validation: Gaussian $\rightarrow$ Bimodal on Unstructured Mesh

2D density transport



Unstructured density transport ($\gamma = 0.05$)

Drift field at $t = 0.56$

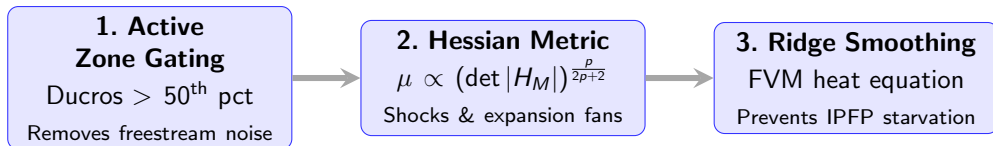


Drift field $b_t = 2\gamma \nabla g_t$ at $t = 0.5$

Application: Feature Extraction Pipeline

Problem

We need marginals μ_0, μ_1 capturing both C^0 discontinuities (shocks) and C^1 discontinuities (expansion fans) on the aerodynamic mesh.

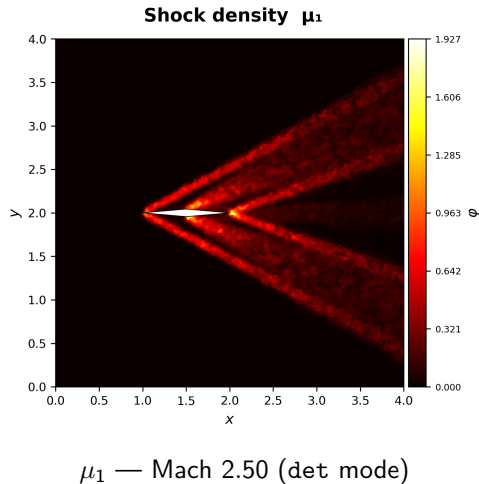
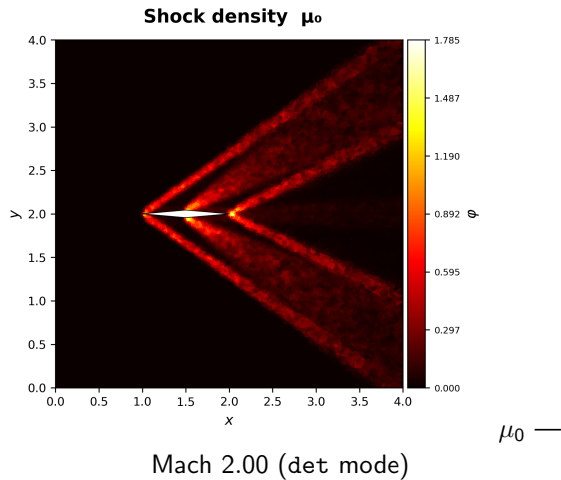


The Alauzet–Loseille metric density (two modes):

$$\det : \quad \mu(x) \propto |\lambda_1 \lambda_2|^{\frac{p}{2p+2}}, \quad \text{eig} : \quad \mu(x) \propto |\lambda_{\max}|^{\frac{p}{2p+2}},$$

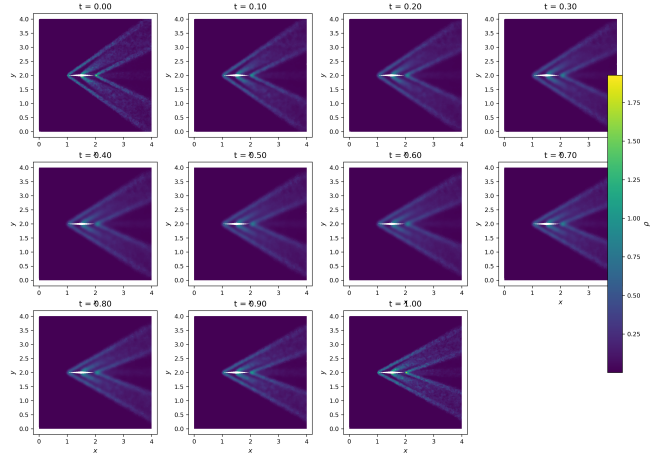
where λ_1, λ_2 are eigenvalues of $|H_M|$, $p = 2$.

Extracted Marginals — Diamond Body (M 2.00 $\rightarrow$ M 2.50)

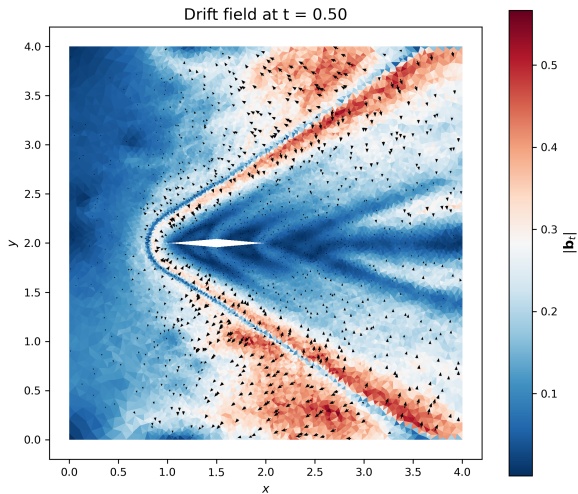


SB Density Transport — Diamond Body

2D density transport



Drift Field $b_t = 2\gamma\nabla g_t$ — Diamond Body



Barycentric SB-CDI: IPFP Potentials $\rightarrow$ Mach Interpolation (1/2)

Step 1 — IPFP $\rightarrow$ Sinkhorn potentials f, g

$$\mu_0 = e^f \cdot \mathcal{P}_1 e^g, \quad \mu_1 = e^g \cdot \mathcal{P}_1 e^f.$$

Step 2 — Barycentric transport maps (boundary-corrected conditional means)

Forward map T (from e^g) and backward map S (from e^f):

$$T(x) = \frac{\mathcal{P}_1[\text{id} \cdot e^g](x)}{\mathcal{P}_1[e^g](x)} - \beta(x), \quad S(x) = \frac{\mathcal{P}_1[\text{id} \cdot e^f](x)}{\mathcal{P}_1[e^f](x)} - \beta(x),$$

where $\beta(x) = \mathcal{P}_1[\text{id}]/\mathcal{P}_1[1] - x$ removes the Neumann identity-leak.

Step 3 — Symmetric two-map CDI reconstruction (explicit, non-iterative)

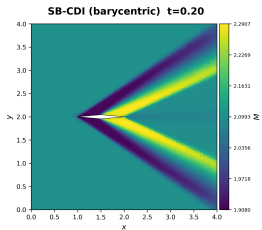
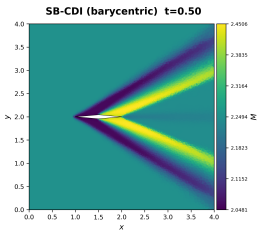
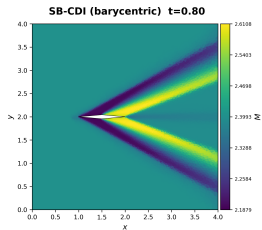
Sample each endpoint field along its own displaced pre-image, using the forward map T ($\mu_0 \rightarrow \mu_1$) and backward map S ($\mu_1 \rightarrow \mu_0$):

$$W(t, x) = (1 - t)x + tS(x), \quad Z(t, x) = tx + (1 - t)T(x),$$

$$\hat{M}_t(x) = (1 - t)M_0(W(t, x)) + tM_1(Z(t, x)).$$

Gradient-free, no inversion, and exact at the endpoints: $\hat{M}_0 = M_0$, $\hat{M}_1 = M_1$.

Mach Interpolation — BaryCDI vs. HR Reference (det mode)

BaryCDI, $t = 0.20$ BaryCDI, $t = 0.50$ BaryCDI, $t = 0.80$

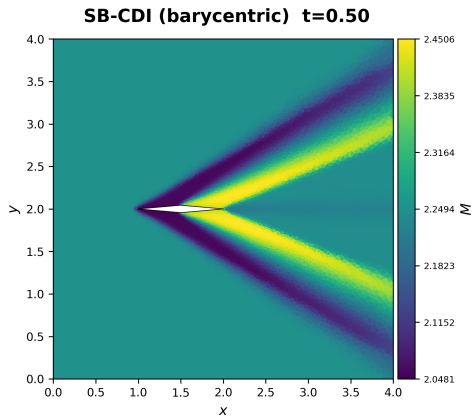
Mach_interpolation/diamondMach/hessran/det/drdiamondMach/hess005n/det/detRef1

HR ref., $M = 2.10$

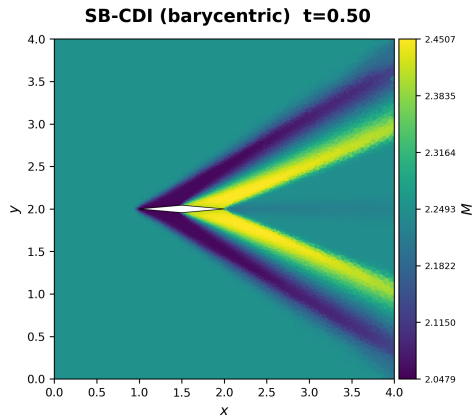
HR ref., $M = 2.25$

HR ref., $M = 2.40$

Mach Interpolation at $t = 0.5$ — det vs eig mode



Barycentric CDI (det mode)



Barycentric CDI (eig mode)

On the diamond's straight oblique shocks, det gives the better, more robust reconstruction; eig is preferred for curved shocks (e.g. the bump).

Quantitative Error: Wasserstein-2 Distance (1/2)

Kantorovich formulation

Given interpolated field $\hat{M}_t$ and reference M_t^{ref} , define $\mu^M(x) \propto |M(x) - M_\infty|$ (normalized to unit mass on the mesh). The W_2 distance:

$$W_2^2(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int_{\Omega \times \Omega} \|x - y\|^2 d\pi(x, y).$$

Entropy-regularised OT cost

With regularisation parameter $\varepsilon = 4\gamma_w$:

$$\text{OT}_\varepsilon(\mu, \nu) = \min_{\pi \in \Pi(\mu, \nu)} \langle C, \pi \rangle - \varepsilon H(\pi), \quad C_{ij} = \|x_i - x_j\|^2.$$

Quantitative Error: Wasserstein-2 Distance (2/2)

Debiased Sinkhorn divergence

Removes the entropic self-bias so that $S_\varepsilon(\mu, \mu) = 0$:

$$S_\varepsilon(\mu, \nu) = \text{OT}_\varepsilon(\mu, \nu) - \frac{1}{2}\text{OT}_\varepsilon(\mu, \mu) - \frac{1}{2}\text{OT}_\varepsilon(\nu, \nu) \approx W_2^2(\mu, \nu).$$

Implementation via convolutional Sinkhorn (Solomon et al. 2015)

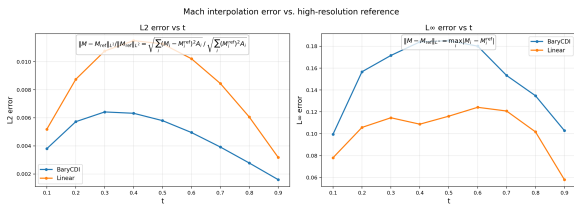
The Gibbs kernel $K_{ij} = e^{-\|x_i - x_j\|^2/\varepsilon}$ is approximated by the FVM heat semigroup $e^{\gamma_w \Delta}$, *the same operator used inside IPFP*.

Dual value at convergence: $\text{OT}_\varepsilon = \varepsilon(\langle F, \mu \rangle + \langle G, \nu \rangle)$, with Sinkhorn potentials F, G satisfying

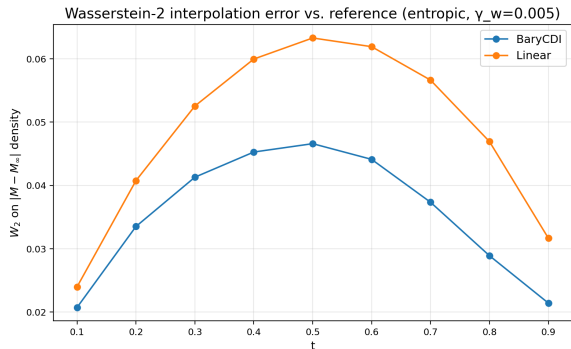
$$G = \log \nu - \log \mathcal{P}_1[e^F], \quad F = \log \mu - \log \mathcal{P}_1[e^G], \quad W_2 \approx \sqrt{\max(S_\varepsilon, 0)}.$$

Parameters: $\gamma_w = 0.005$ ($\sigma_w \approx 0.10$), 50 Sinkhorn iterations.

Error vs. High-Resolution Reference — Diamond (det mode)

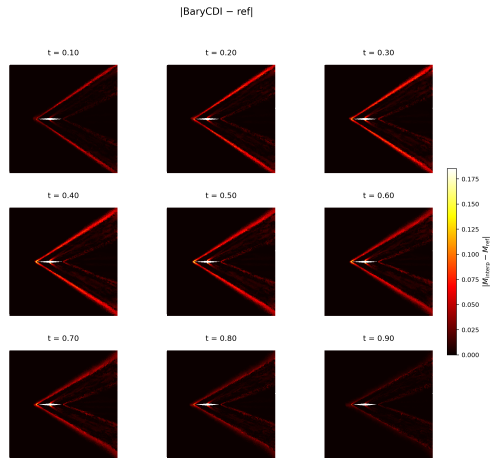
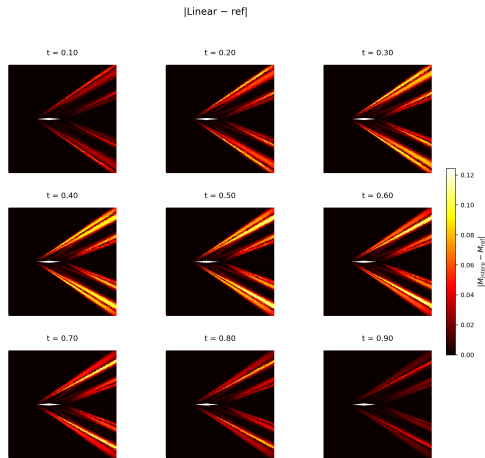


L_2 and L_∞ Mach errors vs. 9 references



W_2 distance on $|M - M_\infty|$ density

Absolute Error Heatmaps — Diamond (det mode)



Analytic Drift for the Diamond Airfoil (1/2) — Shock Geometry

The endpoint marginals live at freestream Mach M_0, M_1 ; the drift $\mathbf{b}(t, \mathbf{x})$ is built to rotate the LE and TE shocks between their M_0 and M_1 wave angles, so the bridge only has to correct residual detail.

Wave angles from the θ - β - M relation

$$\tan \theta = 2 \cot \beta_s \frac{M^2 \sin^2 \beta_s - 1}{M^2(\gamma_g + \cos 2\beta_s) + 2}, \quad \beta_s^{\text{LE}}(M) = \beta_s(\alpha, M).$$

The TE family processes the flow after it crosses the LE shock and expands by 2α around the shoulder (Prandtl–Meyer):

$$\nu(M_b) = \nu(M_a) + 2\alpha, \quad \beta_s^{\text{TE}}(M) = \beta_s(\alpha, M_b(M)).$$

Ray-angle schedule

$$M(t) = (1 - t)M_0 + tM_1, \quad \psi_{\text{LE}}(t) = \beta_s^{\text{LE}}(t), \quad \psi_{\text{TE}}(t) = \beta_s^{\text{TE}}(t) - \alpha, \\ \omega_\kappa(t) = \dot{\psi}_\kappa(t) = \frac{d\beta_s^\kappa}{dM} \Big|_{M(t)} (M_1 - M_0).$$

Analytic Drift for the Diamond Airfoil (2/2) — Windowed Rotation

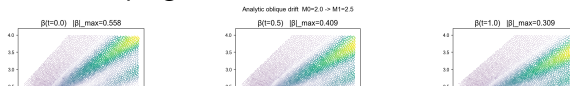
Each ray rotates rigidly about its apex $\mathbf{a}_\kappa$, localized by a Gaussian band of perpendicular width σ_w (widened conically with along-ray distance s to cover the whole swept wedge) and gated downstream:

$$s_{\kappa,\sigma} = (\mathbf{x} - \mathbf{a}_\kappa) \cdot \hat{\mathbf{d}}_{\kappa,\sigma}, \quad w_{\kappa,\sigma} = e^{-\text{dist}_{\kappa,\sigma}^2 / \sigma_w(s)^2} \mathbf{1}[s_{\kappa,\sigma} > 0].$$

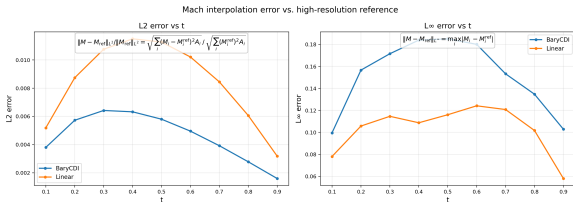
Reference drift

$$\mathbf{b}(t, \mathbf{x}) = \sum_{\kappa \in \{\text{LE}, \text{TE}\}} \sum_{\sigma \in \{+1, -1\}} \sigma \omega_\kappa(t) w_{\kappa,\sigma}(t, \mathbf{x}) \mathbf{R} (\mathbf{x} - \mathbf{a}_\kappa), \quad \mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Vanishes in the freestream; near each shock ray it rotates that shock about its edge from the M_0 angle toward the M_1 angle. Since the localization makes $\nabla \cdot \mathbf{b} \neq 0$, the field enters through the conservative operator $\mathcal{Q}^\dagger$, keeping the reaction term.

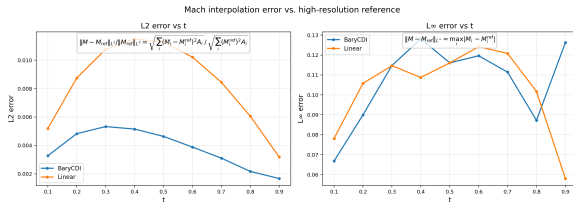


Drifted vs. Driftless Bridge — Error vs. HR Reference (det mode)



No drift, $\gamma_{sb} = 0.005$

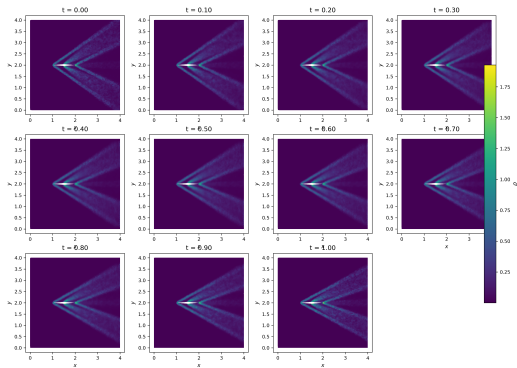
The drift lets IPFP run stably at a $5\times$ smaller γ_{sb} — smaller entropic blur, sharper reconstructed shocks.



Oblique drift, $\gamma_{sb} = 0.001$

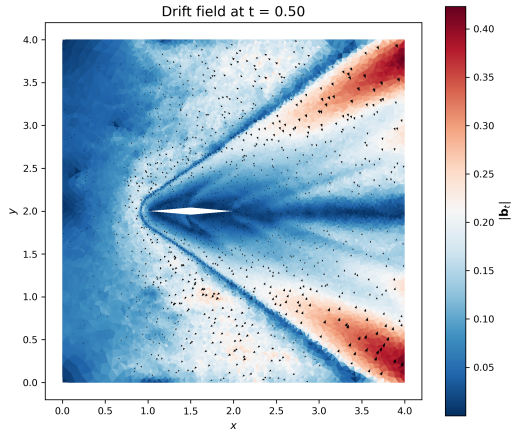
Drifted Bridge — Density Transport and Drift Field (det mode)

2D density transport



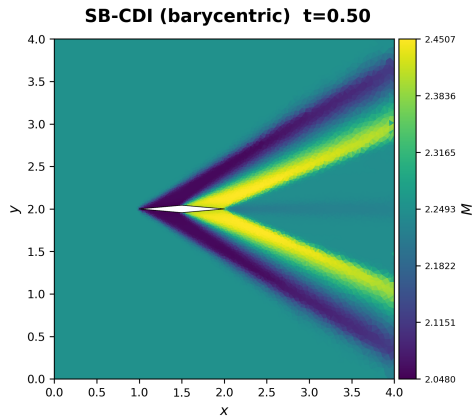
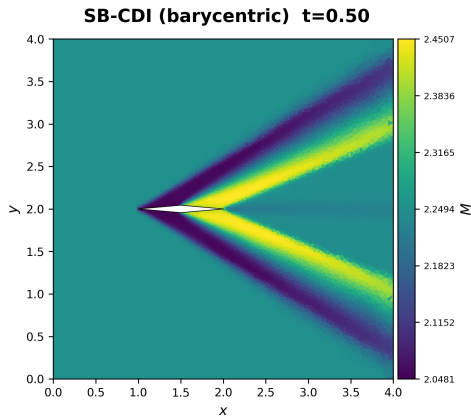
Density transport under $\mathcal{Q}, \mathcal{Q}^\dagger$

Drift field at $t = 0.50$



Drift field $b_t = \mathbf{b}_t + 2\gamma \nabla g_t$ at $t = 0.5$

Mach Interpolation with Oblique Drift — det vs eig ($t = 0.5$)



Barycentric CDI, oblique drift (det)

Barycentric CDI, oblique drift (eig)

Both hessian modes remain stable at $\gamma_{sb} = 0.001$ once the drift pre-transports the shock rays.

Absolute Error Heatmaps — Oblique Drift (det mode, $\gamma_{sb} = 0.001$)

